Group Topic: Urban Air Quality & Environmental Health Lab No. Week 3

Git Repo/ Colab Link: https://github.com/Cristy16/BDA---Final-Project Date: May 7, 2026

Objectives

To develop and train the initial version of the predictive model. The primary goal was to establish a performance benchmark using the "Training Pile" provided by Danilyn, ensuring the chosen algorithm (Random Forest Regressor) could effectively correlate gaseous pollutant levels with PM_2.5 concentrations.

Tasks Completed

1. Model Initialization: Imported the RandomForestRegressor from sklearn.ensemble. Chose this algorithm for its ability to handle non-linear relationships and high-dimensional data resulting from One-Hot Encoding.

2. Baseline Training: Fitted the model using X_train and y_train (the 80% split). Used n_jobs=-1 to optimize training speed across CPU cores.

3. Metric Calculation: Generated predictions on the "Testing Pile" (X_test) to evaluate the model on unseen data.

4. Performance Evaluation: Calculated the Mean Absolute Error (MAE) and R^2 score to quantify the baseline accuracy.

```python
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error, r2_score

# --- 6. Implementation (Baseline) ---
# Training the first version
print("Polly is now training the baseline model... (this may take a minute)")

# Initializing the model with basic settings (n_estimators=100 is standard)
baseline_model = RandomForestRegressor(n_estimators=100, random_state=42, n_jobs=-1)

# Feeding the Training Pile to the model
baseline_model.fit(X_train, y_train)

# --- 7. Initial Prediction Results ---
# We use the X_test (Daryl's pile) to see how this model performs
y_pred = baseline_model.predict(X_test)

mae = mean_absolute_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

print("-" * 30)
print(f"Polly's Baseline Results:")
print(f"Mean Absolute Error: {mae:.4f}")
print(f"R-squared Score: {r2:.4f}")
print("-" * 30)
```

Tasks to be Completed

I was able to do all task assignment during this week 3 activity

Results and Analysis

Metric	Result	Interpretation
Mean Absolute Error (MAE)	5.5402	On average, predictions are within ~5.5 units of the actual PM_2.5 value.
R-squared (R^2)	0.4893	The model explains approximately 49% of the variance in air quality.

The results confirm that CO, NO_2, and SO_2 are strong predictors for PM_2.5. While an R^2 of ~0.49 is a promising start for a baseline, there is a significant margin for improvement through the fine-tuning of forest depth and estimator counts.

Challenges and Solutions

Challenge: RAM issues when training the Random Forest on nearly 400,000 rows.

Solution: Used n_jobs=-1 to distribute the workload and ensured the nrows parameter was balanced during the data loading phase to keep memory usage stable.

Conclusion

The Week 3 objectives were met. The baseline implementation proved that "Prediction" is the correct path for this dataset. The key takeaway was the importance of establishing a benchmark; without this baseline, the team would have no way to measure the success of the optimization steps following in the next phase.